

LaTeX Scribe Generation Request

I need to create a LaTeX document based on the content of the PDF file "L18.pdf." The doc

I have also provided a sample scribe titled "s1_g6_ITS_L6_Scribe_1_Shot_AU2440052_Print

Based on the content of **L18.pdf**, here is the LaTeX scribe document. It follows the structure of your provided sample, incorporating a

Code snippet

```
\documentclass[11pt]{article}
\usepackage[utf8]{inputenc}
\usepackage{amsmath, amssymb, amsfonts, amsthm}
\usepackage{geometry}
\usepackage{graphicx}
\usepackage{hyperref}

\geometry{a4paper, margin=1in}

\begin{document}

% --- Header Section ---
\begin{center}
\textbf{\Large CSE400: Fundamentals of Probability in Computing} \\
\textbf{Lecture 18: Marginal PMF, Correlation, Covariance, Joint Gaussian RVs} \\
\textit{Scribe for Lecture held on March 12, 2026} [cite: 8, 11] \\
\textit{Instructor: Dhaval Patel, PhD} [cite: 3, 9]
\end{center}

\hrule
\vspace{0.5cm}

% --- Section 1: Marginal PMF ---
\section{Marginal Probability Mass Function (PMF)}
For discrete random variables  $X$  and  $Y$ , the joint behavior is described by the joint PMF  $p_{XY}(x,y) = P(X=x,$ 

\subsection{Definition}
The marginal PMF of one variable is obtained by summing the joint PMF over all possible values of the other varia
\begin{itemize}
\item  $p_X(x) = \sum_y p_{XY}(x,y)$  [cite: 75, 76]
\item  $p_Y(y) = \sum_x p_{XY}(x,y)$  [cite: 76]
\end{itemize}

\subsection{Marginal PMF from a Joint PMF Table}
In a tabular representation, the row sums provide  $p_X(x)$ , and the column sums provide  $p_Y(y)$  [cite: 85, 93]
\begin{itemize}
\item  $p_X(x_{i1}) = \sum_j p_{ij}$  and  $\sum_i p_{ij} = 1$  [cite: 96]
\item  $p_Y(y_{j1}) = \sum_i p_{ij}$  and  $\sum_j p_{ij} = 1$  [cite: 97, 98]
\end{itemize}

% --- Section 2: Correlation ---
\section{Correlation}
Correlation is a basic way to measure the joint behavior of two random variables through the "product moment" [cit

\subsection{Definition}
In the continuous case, the raw correlation  $R_{XY}$  is defined as [cite: 135, 136]:

$$R_{XY} = E[XY] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{XY}(x,y) dx dy$$
 [cite: 137]
\begin{itemize}
\item It represents the average value of the product  $XY$  [cite: 138].
\item If  $R_{XY} = 0$ , the variables are considered orthogonal about the origin [cite: 150, 151].
\item \textbf{Limitation:} It depends on the scale of  $X$  and  $Y$  and mixes mean values, spread, and dependen
\end{itemize}
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% --- Section 3: Covariance ---
\section{Covariance}
To create a cleaner measure of dependence, we center the variables by subtracting their means[cite: 152, 154].

\subsection{Definition and Interpretation}
Covariance measures how centered quantities vary together[cite: 165]:

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$
 [cite: 162]
where  $\mu_X = E[X]$  and  $\mu_Y = E[Y]$  [cite: 163].
\begin{itemize}
\item \textbf{Positive Covariance} ( $>0$ ): Deviations tend to have the same sign (e.g., as study hours increase, grades tend to increase).
\item \textbf{Negative Covariance} ( $<0$ ): One variable increases while the other decreases (e.g., as TV time increases, study hours tend to decrease).
\item \textbf{Zero Covariance} ( $=0$ ): No linear co-variation trend[cite: 174, 177].
\end{itemize}

\subsection{Algebraic Form and Properties}
The computational formula for covariance is[cite: 216, 227]:

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$
 [cite: 229]

\textbf{Key Properties:}
\begin{enumerate}
\item \textbf{Symmetry:}  $\text{Cov}(X, Y) = \text{Cov}(Y, X)$  [cite: 235].
\item \textbf{Variance as Special Case:}  $\text{Cov}(X, X) = \text{Var}(X)$  [cite: 244, 253].
\item \textbf{Linearity:}  $\text{Cov}(aX + b, cY + d) = ac \text{Cov}(X, Y)$  [cite: 255, 265].
\item \textbf{Independence:}  $X \perp Y \Rightarrow \text{Cov}(X, Y) = 0$  (Independence implies zero covariance, but not vice versa).
\end{enumerate}

% --- Section 4: Pearson's Coefficient ---
\section{Pearson's Correlation Coefficient ( $\rho_{XY}$ )}
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The Pearson coefficient is the normalized version of covariance[cite: 353, 386].

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\subsection{Definition}

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$
 [cite: 355, 384, 392]
\begin{itemize}
\item \textbf{Range:} Always between -1 and 1 ( $-1 \leq \rho_{XY} \leq 1$ ) [cite: 354, 397].
\item \textbf{Unit-free:} It is independent of the units of  $X$  and  $Y$  [cite: 356, 387].
\item \textbf{Strength:} Unlike covariance, it measures both the direction and the \textit{strength} of the linear relationship.
\end{itemize}

% --- Section 5: Jointly Gaussian RVs ---
\section{Jointly Gaussian Random Variables}
A pair  $(X, Y)$  is jointly Gaussian if its joint density has a bivariate Gaussian form and the marginals are Gaussian.

\subsection{Joint PDF Form}
For two variables  $X$  and  $Y$  [cite: 589]:

$$f_{X,Y}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left[\frac{x^2}{\sigma_X^2} - 2\rho\frac{xy}{\sigma_X\sigma_Y} + \frac{y^2}{\sigma_Y^2}\right]\right)$$


\subsection{Applications}
\begin{itemize}
\item \textbf{Signal Processing & Communication:} Natural model for continuous noisy measurements[cite: 580, 581].
\item \textbf{Climate Modeling:} Used to study relationships like  $\text{CO}_2$  concentration and temperature anomalies.
\end{itemize}

% --- Exam Revision Summary ---
\section*{Summary for Exam Revision}
\begin{table}[h]
\centering
\begin{tabular}{|l|l|}
\hline
\textbf{Concept} & \textbf{Key Formula / takeaway} \\ \hline
Marginal PMF & Sum the joint PMF over the "other" variable. [cite: 78] \\ \hline
Correlation ( $\rho_{XY}$ ) &  $E[XY] - E[X]E[Y]$ . Measures raw joint behavior. [cite: 136] \\ \hline
Covariance ( $\text{Cov}$ ) &  $E[XY] - E[X]E[Y]$ . Measures linear direction. [cite: 227] \\ \hline
Pearson's ( $\rho$ ) &  $\frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$ . Measures direction AND strength (-1 to 1). [cite: 355] \\ \hline
Independence & If independent,  $\text{Cov}=0$  and  $\rho=0$ . [cite: 268, 558] \\ \hline
Joint Gaussian & Marginals are Gaussian; defined by  $\mu, \sigma, \rho$ . [cite: 568, 649] \\ \hline
\end{tabular}
\end{table}

\end{document}

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Would you like me to generate a set of practice problems based on the solved examples in this lecture?

